Tecnológico Nacional de México
Campus Cd. Madero.

Ingeniería en Electrónica.

Programación Estructurada.

Evaluación del curso.

- Individual
- Proyectos y prácticas 100%
    - Se suben a Github (Repositorios privados y con invitación)
    - El trabajo se evalúa en base a cumplimiento de requerimientos
    - Posteriormente hay una evaluación personal y puede modificar la nota del trabajo

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Prácticas:

1. Radar Speed Traps

A highway-patrol speed-detection radar emits a beam of microwaves at a frequency designated as f_e . The beam is reflected off an approaching car, and the radar unit picks up and analyzes the reflected beam's frequency, f_r. The reflected beam's frequency is shifted slightly from f_e to f_r because of the car's motion. The relationship between the speed of the car, v, in miles per hour (mph), and the two microwave frequencies is:

$$v = (6.685 \times 10^8)\left(\frac{f_r - f_e}{f_r + f_e}\right) mph$$

where the emitted waves have a frequency of f_e = 2 × 1010 sec^{-1}. Using the given formula, you'll write a C++ program, using the software development procedure, to calculate and display the speed corresponding to a given received frequency f_r.

2. Temperature conversion

Write a C++ program that accepts a number followed by one space and then a letter. If the letter following the number is f, the program is to treat the number entered as a temperature in degrees Fahrenheit, convert the number to the equivalent degrees Celsius, and display a suitable message. If the letter following the number is c, the program is to treat the number entered as a temperature in degrees Celsius, convert the number to the equivalent degrees Fahrenheit, and display a suitable message. If the letter is neither f nor c, the program is to display a message that the data entered is incorrect and terminate. Use an if-else chain in your program and make use of these conversion formulas:

Celsius = (5.0 / 9.0) × (Fahrenheit - 32.0)

Fahrenheit = (9.0 / 5.0) × Celsius + 32.0